Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

[a4paper,10pt]article fontspec eurosym

Mapping=tex-text Fontin-Regular.otf[BoldFont = Fontin-Bold.otf , ItalicFont = Fontin-Italic.otf , SmallCapsFont = Fontin-SmallCaps.otf]

xunicode,xltxtra,url,parskip [usenames,dvipsnames]xcolor [big]layaureo hyperref linkcolourrgb0,0.2,0.6 titlesec

0em[]

0pt3pt3pt

Dr. Martin Hölzer

1 Personal Data

Name:	Dr. rer. nat. Martin Hölzer
Gender:	Male
Place and Date of Birth:	Rudolstadt, Germany | 16 March 1988
Address:	Jansonstr. 11, 07745 Jena
Phone:	+49 160 92675865
email:	martin.hoelzer@uni-jena.de
Family status:	Single

2 Work Experience

since 02/2018	Post-Doc at RNA Bioinformatics/High-Throughput Analysis, Prof. Dr. Manja Marz, University of Jena
since 03/2017	Founding member of the "European Virus Bioinformatics Center" (EVBC), University of Jena
08/2013–06/2017	PhD student, RNA Bioinformatics/High-Throughput Analysis, Prof. Dr. Manja Marz, University of Jena

3 Scientific Education

01/2018	Doctorate degree, Dr. rer. nat. University of Jena Grade: summa cum laude
08/2013–06/2017	PhD student at University of Jena
12/2012	Diploma in Bioinformatics University of Jena Grade: 1.4
10/2007–12/2012	Studies of Bioinformatics at University of Jena

4 Ten Publications

P. Möbius, G. Nordsiek, **M. Hölzer**, M. Jarek, M. Marz, and H. Köhler. Complete Genome Sequence of JII-1961, a Bovine *Mycobacterium avium* subsp. *paratuberculosis* Field Isolate from Germany. *Genome Announc.* 5(34), 2017.

J. Fuchs, **M. Hölzer**, M.Schilling, C. Patzina, A. Schoen, T. Hoenen, G. Zimmer, M. Marz, F. Weber, M. A. Müller, and G. Kochs. Evolution and Antiviral Specificities of Interferon-Induced Mx Proteins of Bats against Ebola, Influenza, and Other RNA Viruses. *J Virol.* 12; 91(15), 2017.

M. Hölzer, V. Krähling, F. Amman, E. Barth, S. H. Bernhart, V. A. O. Carmelo, M. Collatz, G. Doose, F. Eggenhofer, J. Ewald, J. Fallmann, L. M. Feldhahn, M. Fricke, J. Gebauer, A. J. Gruber, F. Hufsky, H. Indrischek, S. Kanton, J. Linde, N. M. Berrospi, R. Ochsenreiter, K. Rieger, L. Rivarola-Duarte, A. H. Sahyoun, S. J. Saunders, S. E. Seemann, A. Tanzer, B. Vogel, S. Wehner, M. T. Wolfinger, R. Backofen, J. Gorodkin, I. Grosse, I. Hofacker, S. Hoffmann, C. Kaleta, P. F. Stadler, S. Becker, and M. Marz. Differential transcriptional responses to Ebola and Marburg virus infection in bat and human cells. *Scientific reports* 6:34589, 2016.

A. H. Sahyoun, **M. Hölzer**, F. Jühling, C. Höner Zu Siederdissen, M. Al-Arab, K. Tout, M. Marz, M. Middendorf, P. F. Stadler, and M. Bernt. Towards a comprehensive picture of alloacceptor tRNA remolding in metazoan mitochondrial genomes. *Nucleic Acids Res.* 43(16):8044–8056, 2015.